

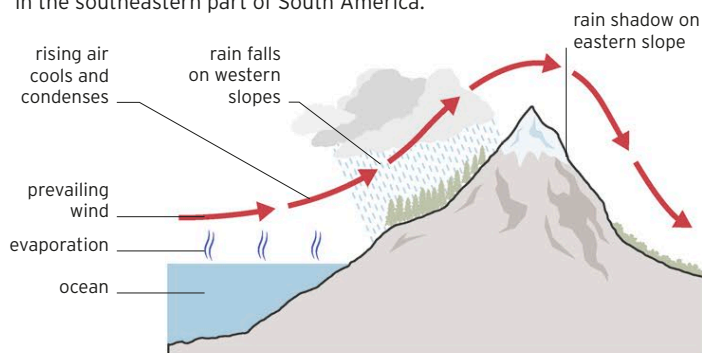


#### △ DIRTY THUNDERSTORM

Lightning in and around a volcanic ash plume—as in this 2015 eruption of the Calbuco volcano in Chile—is referred to as a dirty thunderstorm. The flashes of lightning result from friction between ash particles and the atmosphere.

#### THE SOUTHERN ANDES RAIN SHADOW

In the southern Andes, the prevailing winds blow from the west and push moisture-laden air from the Pacific Ocean toward the mountains. As this air meets the Andes, it rises, expands, and cools, dropping rain on the mountains' western flanks. However, it then continues over and down the eastern side as a dry wind, creating arid conditions (a rain shadow) in the southeastern part of South America.



#### △ ROOF OF THE AMERICAS

With its highest point at 22,838 ft (6,961 m), Mount Aconcagua, seen here at the center, is the highest peak not just in the Andes but in the entire world outside of Asia. It lies in Argentina, just east of the country's border with Chile.

Cotopaxi, in Ecuador, is one of the **highest active volcanoes** in the world. It has **erupted 50 times** since **1738**